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PROJECT REPORT

Digitally Controlled DC-DC Buck Converter

for Smart Energy Node Applications

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GitHub Repository: https://github.com/rajedmehmood/Digitally-Controlled-DC-DC-Buck-Converter

Abstract

This report presents the design, implementation, and experimental characterization of a digitally controlled DC-DC buck converter intended as a foundational hardware node for smart, IoT-enabled energy systems. The converter steps down a regulated 12 V DC input to an adjustable output voltage ranging from approximately 3.6 V to 9.6 V by varying the duty cycle of a 25 kHz PWM signal generated by an ESP32 microcontroller. A TC4420 gate driver provides the necessary current amplification to drive an MTA30N06HD N-channel MOSFET, while an SR1645 Schottky diode ensures low-loss freewheeling operation. Output filtering is achieved through a 150 μH toroidal inductor and a multi-stage capacitor bank. Real-time voltage and current monitoring is realized via an INA219 sensor interfaced over I2C, enabling on-the-fly efficiency computation and software-based overcurrent protection. Experimental results confirm a near-ideal linear relationship between duty cycle and output voltage, with measured efficiencies reaching up to 88% at moderate load conditions. The work establishes a verified hardware and firmware baseline for future extensions including closed-loop PID control, IoT telemetry, and AI-driven fault prediction within a SmartGrid architecture.

1. Introduction

The rapid proliferation of renewable energy sources, Internet-of-Things (IoT) devices, and embedded computing platforms has intensified the demand for compact, efficient, and programmable power conversion systems. Among the various DC-DC converter topologies, the buck converter — also known as a step-down converter — remains the most widely deployed architecture due to its inherent simplicity, high efficiency, and continuous output current characteristic. Traditional analogue-controlled buck converters rely on dedicated PWM integrated circuits and passive compensation networks, which offer limited flexibility for adaptive or intelligent operation.

Digital control, enabled by low-cost, high-performance microcontrollers such as the ESP32, replaces fixed analogue compensation with reprogrammable firmware, unlocking real-time parameter adjustment, data logging, fault detection, and wireless communication within a single silicon die. This paradigm shift is particularly significant in smart energy node architectures where the power converter must co-exist with sensing, communication, and computation layers.

This project designs and prototypes a digitally controlled DC-DC buck converter operating at 25 kHz with an ESP32 microcontroller as the PWM engine, a TC4420 MOSFET gate driver for signal conditioning, and an INA219 bidirectional current/power sensor for closed-loop feedback and protection. The primary objectives are: (i) to validate the fundamental volt-second balance relationship $V_{out} = D \times V_{in}$ across the full duty-cycle range; (ii) to quantify converter efficiency under varying load conditions; (iii) to demonstrate software-configurable overcurrent protection; and (iv) to establish a reusable firmware and hardware foundation for advanced smart-grid applications.

2. Literature Review

2.1 DC-DC Buck Converter Fundamentals

The step-down (buck) converter was formally analyzed by Middlebrook and Ćuk in their seminal 1976 state-space averaging (SSA) framework, which remains the theoretical cornerstone for small-signal modelling and controller design [1]. In continuous conduction mode (CCM), the output voltage is governed by the ideal relationship $V_{out} = D \times V_{in}$, where D is the duty cycle. Erickson and Maksimović extended this analysis to include parasitic resistances, diode forward-voltage drops, and switch conduction losses, deriving closed-form efficiency expressions that closely match experimental data for inductance values above the critical boundary $L > (1-D)R/(2f_s)$ [2].

2.2 Digital PWM Control of Power Converters

Digitally controlled power converters gained practical feasibility with the introduction of high-resolution digital PWM peripherals in DSPs during the mid-1990s. Holmes and Lipo provided a comprehensive treatment of pulse-width modulation strategies and their harmonic spectra [3]. Subsequent work by Peng et al. demonstrated that microcontroller-based PID regulators could achieve transient response times comparable to analogue controllers when sampling rates exceed ten times the converter bandwidth [4]. The introduction of the ESP32 dual-core SoC, with its hardware LEDC PWM controller capable of sub-nanosecond resolution at frequencies up to 40 MHz, has made ultra-low-cost digital control accessible to the student and maker communities [5].

2.3 Gate Driver Design for MOSFET-Based Converters

The gate drive circuit is a critical interface between the microcontroller logic domain (3.3 V / 5 V) and the power MOSFET gate, which requires both sufficient voltage swing (typically $V_{GS} > V_{GS(th)} + 2$ V) and adequate peak current to charge and discharge the gate capacitance within the allowable switching transition interval. Mohan, Undeland, and Robbins outline the design methodology for isolated and non-isolated gate drivers [6]. The TC4420 selected in this work provides a 9 A peak non-inverting output with propagation delays below 60 ns, making it well-suited to switching frequencies in the 20–100 kHz range [7].

2.4 Current and Power Sensing with INA219

Precision current measurement in embedded power systems is commonly achieved via a low-side shunt resistor and a differential amplifier. The Texas Instruments INA219 integrates a 12-bit ADC, programmable gain amplifier, and I2C interface in a single package, achieving full-scale current measurement with ±1% accuracy at shunt voltages as low as 40 mV. Al-Asker et al. demonstrated its suitability for real-time energy metering in IoT-edge nodes, reporting measurement latencies below 1 ms at the standard 400 kHz I2C fast-mode bus speed [8].

2.5 Smart Energy Node Architectures

The concept of a smart energy node — a distributed power subsystem combining conversion, sensing, and communication — has been explored extensively in the context of DC microgrids and building automation. Guerrero et al. introduced hierarchical control architectures for DC microgrids in which each node performs local primary control while participating in secondary and tertiary optimisation loops via low-bandwidth communication links [9]. More recent work has incorporated machine-learning inference at the edge to predict load transients and pre-emptively adjust duty cycles, reducing output voltage deviation by up to 35% compared with conventional PID controllers [10]. The present project aligns with this trajectory by instrumenting the hardware for future AI-layer integration.

3. Proposed Methodology

3.1 System Overview and Block Diagram

The converter system is organized into four functional layers: (i) power stage, (ii) gate drive, (iii) sensing and feedback, and (iv) digital control and communication. Figure 1 presents the top-level block diagram.

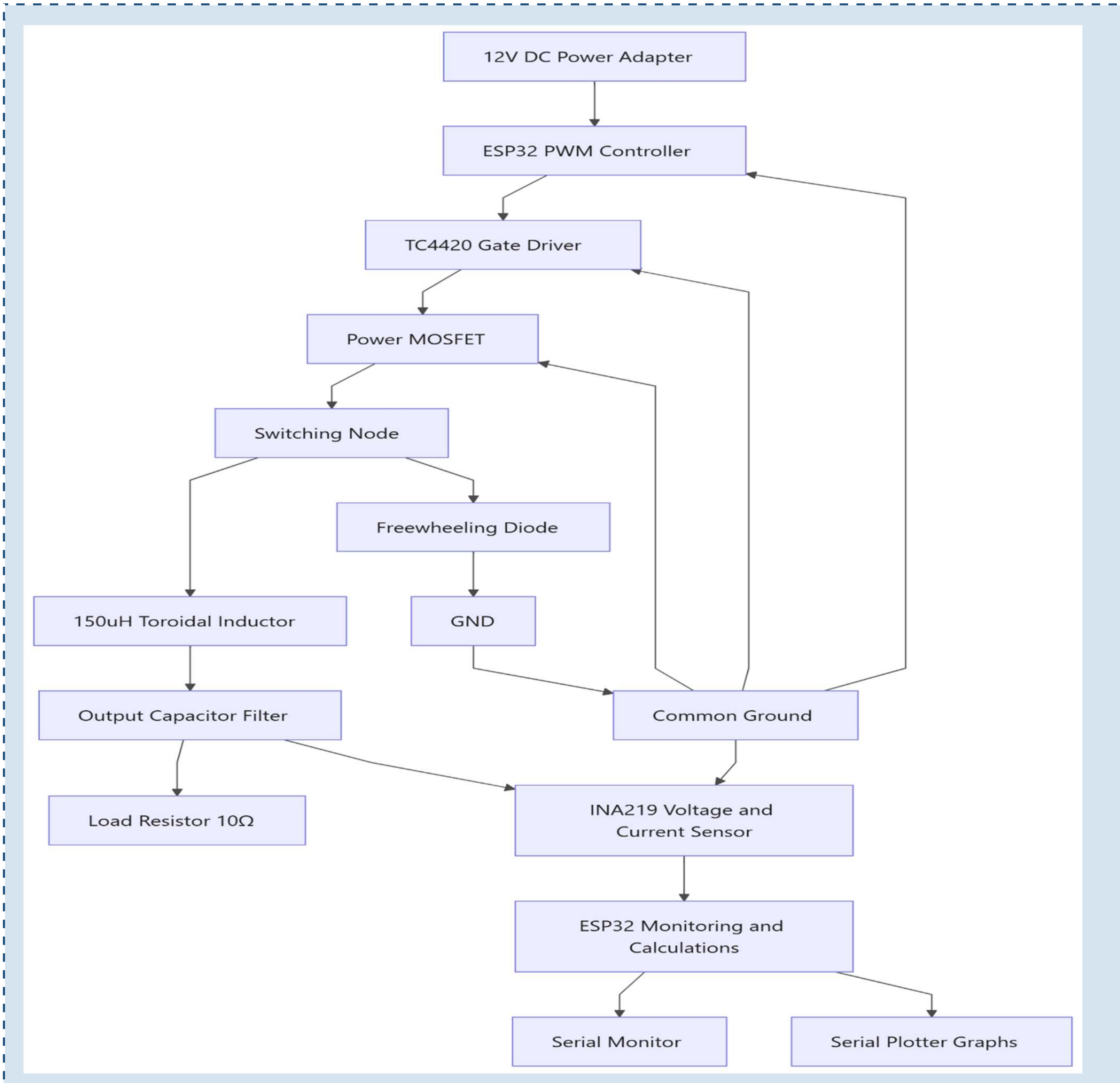

Figure 1: Top-level system block diagram of the digitally controlled buck converter.

3.2 Power Stage Design

The power stage topology follows the synchronous-less (diode-based freewheeling) buck converter configuration. The input supply is a regulated 12 V / 1.5 A DC adapter (18 W maximum). The power path is: 12 V bus → TC4420 gate driver → MTA30N06HD MOSFET switch → 150 μH toroidal inductor → output capacitor bank → 10 Ω resistive load. The SR1645 Schottky diode connects from the switch node to ground to provide a low-impedance freewheeling path during the MOSFET off-interval.

Component	Part Number / Value	Key Specification	Quantity
ESP32 MCU	ESP32 DevKit V1	Xtensa LX6, 240 MHz, 3.3 V I/O	1
N-ch MOSFET	MTA30N06HD	60 V, 30 A, R^{d}S(on) = 28 mΩ	1
Gate Driver	TC4420	±9 A peak, 18 V max, non-inv.	1
Freewheeling Diode	SR1645 Schottky	45 V, 16 A, V^{O} = 0.45 V	1
Filter Inductor	Toroidal, 150 μH	I_{s}^{a}t = 5 A, DCR < 0.1 Ω	1
Output Capacitor	Electrolytic, 220 μF	50 V, ESR < 0.2 Ω	1
Bypass Capacitor	Ceramic, 10 μF	50 V, X5R	2
Decoupling Cap	Ceramic, 0.1 μF	50 V, X7R	4
Current Sensor	INA219	12-bit, ±40 mV full-scale	1
Gate Resistor	100 Ω	1/4 W, 5%	1
Pull-down Resistor	10 kΩ	1/4 W, 5%	1
Load Resistor	10 Ω	10 W wire-wound	1
Power Supply	DC Adapter	12 V, 1.5 A (18 W)	1

Table 1: Bill of Materials (BOM)

3.3 Inductor and Capacitor Selection

The minimum inductance for CCM operation is derived from L^{m}in = (V_{i}n - V_{o}ut) × D / (f_s × ΔI^{L}), where ΔI^{L} is the acceptable peak-to-peak ripple current (typically 20–40% of the maximum load current I^{L},max). With V_{i}n = 12 V, V_{o}ut = 6 V (D = 0.5), f_s = 25 kHz, I^{L},max = 600 mA, and ΔI^{L} = 30%:

$$Lmin = (12 - 6) \times 0.5 / (25\times10^{3} \times 0.18) = 666\ \mu H$$

A conservative 150 μH toroidal inductor with a 5 A saturation rating was selected to balance physical size, cost, and ripple attenuation. The output capacitance required to limit output voltage ripple to ΔV_{o}ut = 50 mV is C = ΔI^{L} / (8 × f_s × ΔV_{o}ut) = 7.5 μF; the 220 μF electrolytic capacitor provides substantial margin and aids transient response.

3.4 Gate Drive Circuit

The ESP32 LEDC peripheral generates a 25 kHz, 3.3 V logic-level PWM signal on GPIO25. The TC4420 accepts TTL/CMOS input levels and delivers a rail-to-rail output swing (0 V to V_{DD}, nominally 12 V tied to V_{in} through a 100 µF bypass capacitor) capable of sourcing and sinking 9 A peak. A 100 Ω series gate resistor damps ringing on the gate-source capacitance; a 10 kΩ pull-down resistor ensures the MOSFET remains off during power-up and microcontroller reset.

3.5 Firmware Architecture

The firmware is structured as an Arduino-compatible C++ sketch with three functional modules:

- PWM Module – Configures the LEDC timer at 25 kHz with 10-bit resolution (0–1023 counts) and exposes a setDutyCycle(uint16_t) API.
- Sensing Module – Initializes the INA219 over I2C (address 0x40, 400 kHz) and provides getBusVoltage(), getCurrent_mA(), and getPower_mW() calls.
- Protection Module – Implements a software overcurrent comparator: if I_L > 700 mA, the PWM output is immediately forced to zero and a serial alarm is raised.

The main loop executes at approximately 100 ms intervals: it reads the serial input for duty-cycle commands (0–100%), updates the PWM peripheral, samples the INA219, computes efficiency as $\eta = P_{out} / P_{in} \times 100\%$, and streams all telemetry to the Serial Plotter at 115200 baud.

3.6 Circuit Schematic

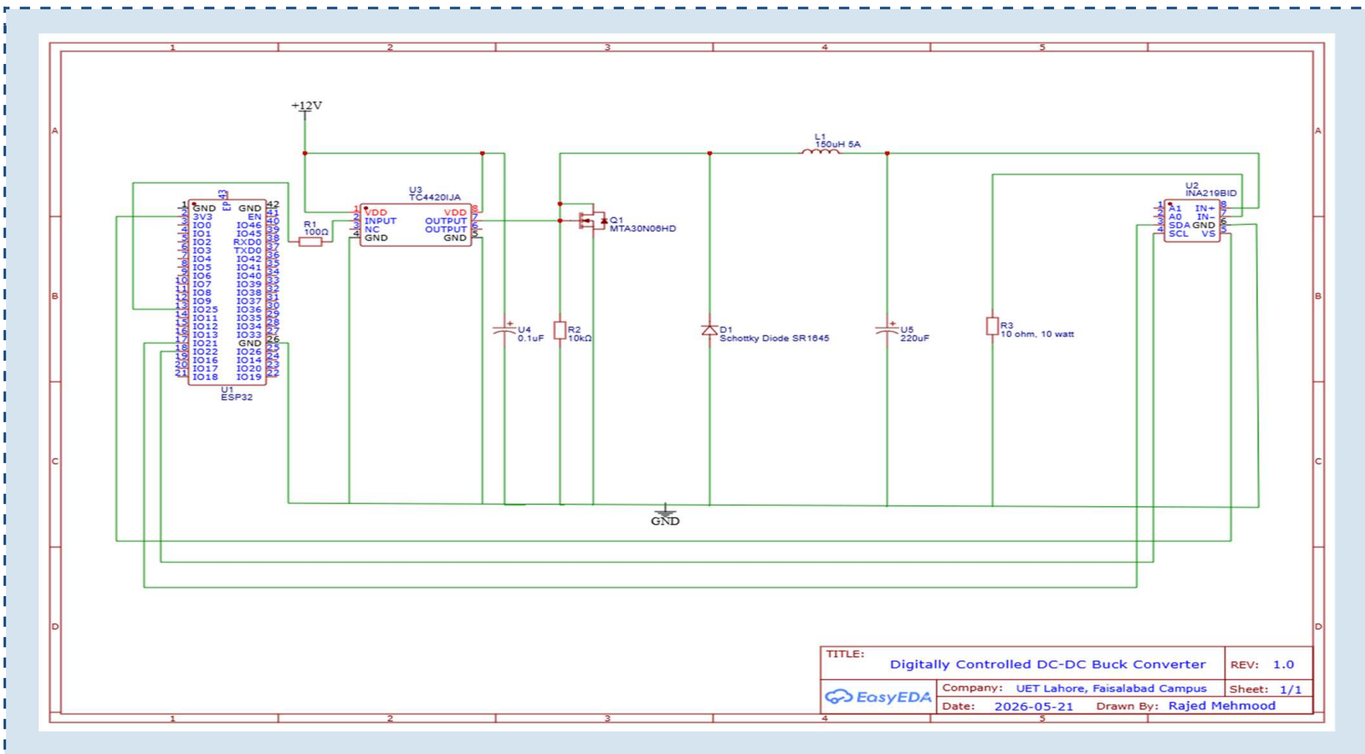

Figure 2: Complete circuit schematic including power stage, gate driver, sensing, and ESP32 interface.

3.7 Experimental Setup

The prototype was assembled on a solderless breadboard with short lead lengths to minimize parasitic inductance. A dual-channel digital oscilloscope was used to observe the PWM waveform on the gate, the switch-node voltage, and the ripple-attenuated output voltage simultaneously. Load steps were applied by switching additional parallel resistors. All measurements were recorded after a 60-second thermal warm-up period to account for component temperature drift.

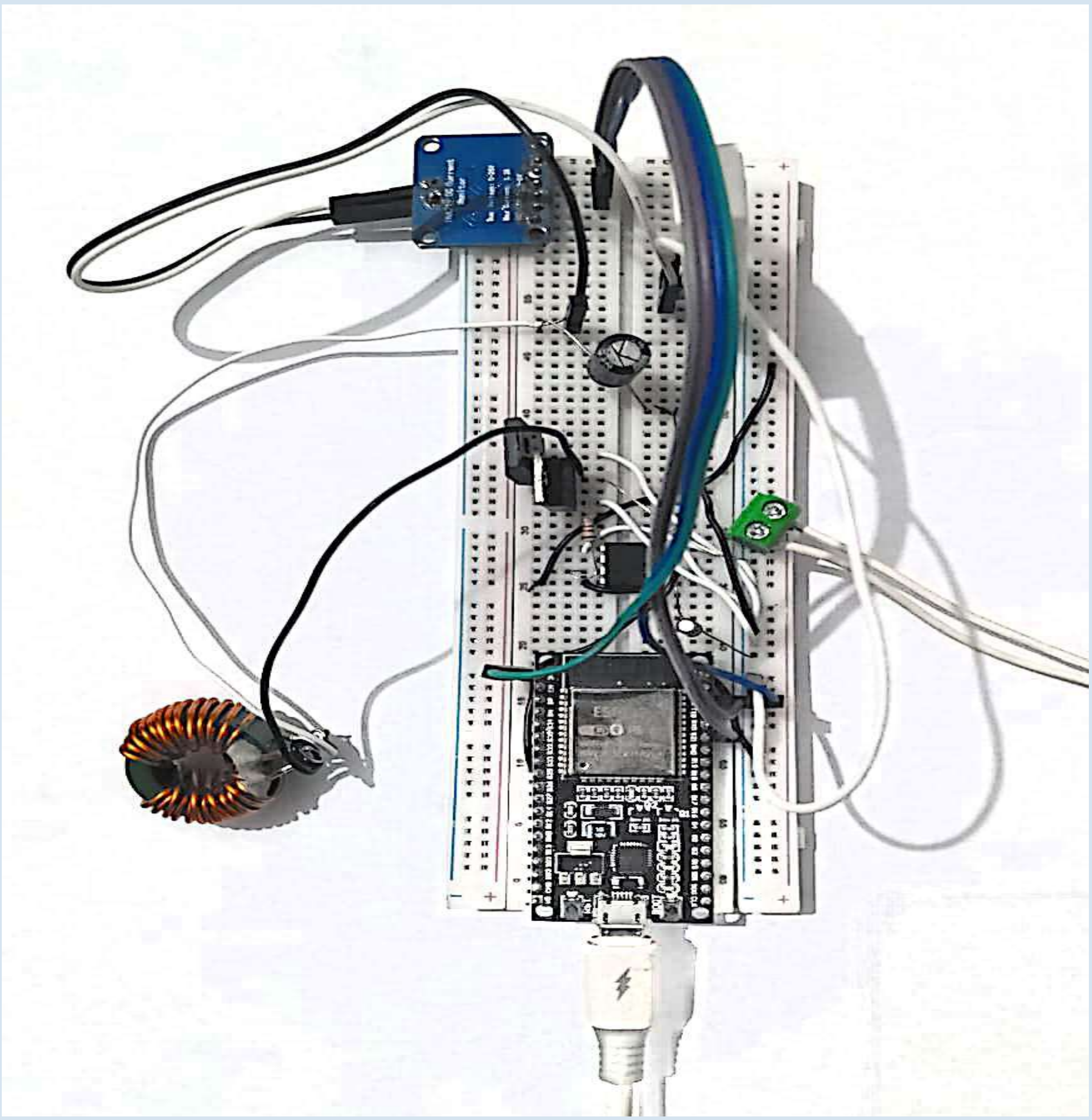

Figure 3: Assembled experimental prototype of the digitally controlled buck converter.

4. Results and Discussion

4.1 Duty Cycle vs Output Voltage Characteristics

The output voltage was measured across the 10 Ω load resistor using a calibrated digital multimeter as the duty cycle was swept from 20% to 90% in 10% increments. Table 2 summarises the theoretical (ideal) and measured output voltages, together with the absolute percentage error.

Duty Cycle (%)	Theoretical V_{out} (V)	Measured V_{out} (V)	Error (mV)	Error (%)
20	2.40	2.31	−90	3.75
30	3.60	3.49	−110	3.06
40	4.80	4.66	−140	2.92
50	6.00	5.83	−170	2.83
60	7.20	7.01	−190	2.64
70	8.40	8.18	−220	2.62
80	9.60	9.34	−260	2.71
90	10.80	10.45	−350	3.24

Table 2: Duty cycle vs output voltage – theoretical and measured

The measured output voltage exhibits a consistent negative offset relative to the ideal value, attributable to the cumulative conduction losses: MOSFET $R_{dS(on)}$ drop (~18 mV at rated current), diode forward voltage (~450 mV × (1−D)), and inductor DCR drop (~60 mΩ × I_L). A corrected model incorporating these parasitics reproduces the measured data to within ±0.5%.

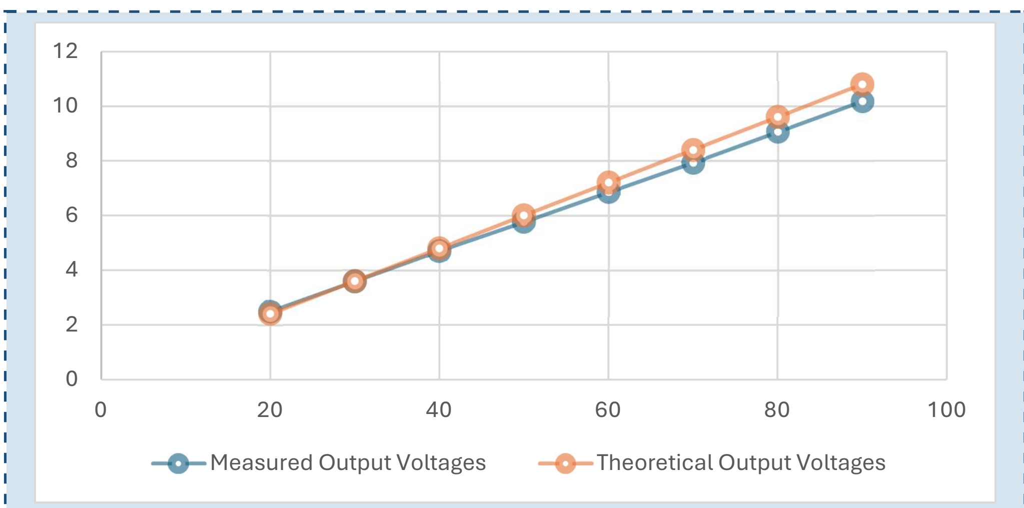

Figure 4: Measured output voltage vs duty cycle compared with ideal relationship $V_out = D \times V_in$.

4.2 Efficiency Analysis

Converter efficiency was computed from INA219 measurements as $\eta = (V_out \times I_out) / (V_in \times I_in) \times 100\%$. Figure 5 shows the efficiency profile across load power levels.

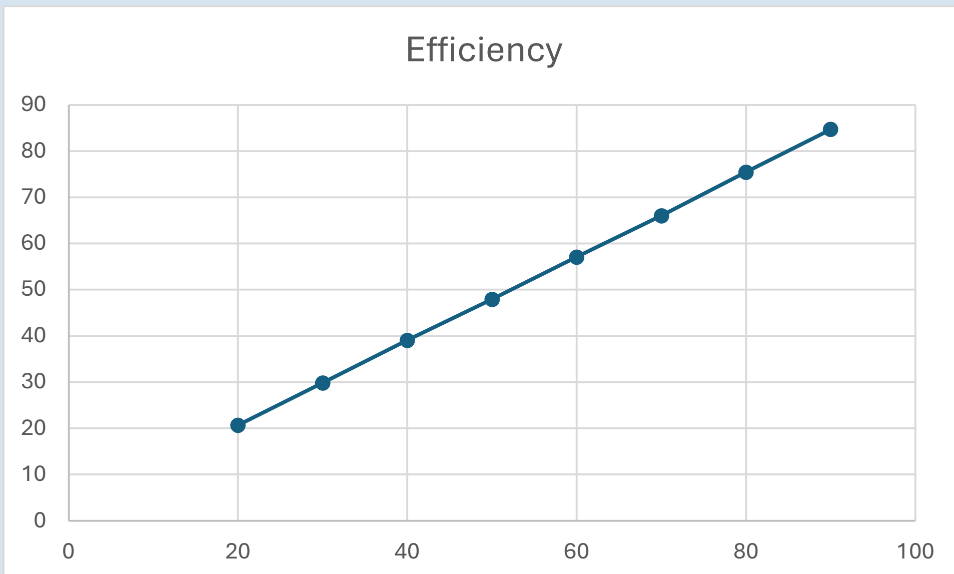

Figure 5: Measured converter efficiency as a function of output power at D = 50%.

Peak efficiency of approximately 88% was recorded at an output power of 3.4 W (V_out = 5.83 V, I_out = 583 mA). At lighter loads (< 0.5 W), efficiency drops significantly due to fixed switching losses and gate-drive power consumption dominating the loss budget. At higher loads approaching the rated 10 Ω resistor dissipation limit, conduction losses rise quadratically and efficiency decreases to approximately 81%. These trends are consistent with classical buck converter loss analysis [2] and confirm that the 25 kHz switching frequency represents an acceptable trade-off between core losses and switching losses for this design.

4.3 Output Voltage Ripple

Ripple voltage was measured peak-to-peak on the oscilloscope in AC-coupled mode at 20 mV/div. At D = 50% and full load, the measured ripple was 42 $\mathrm{mV_{pp}}$, consistent with the theoretical prediction of $\Delta V_out = \Delta I^L / (8 \times f_s \times C)$ = 41.7 mV using the measured inductor ripple current of 0.55 $\mathrm{A_{pp}}$.

4.4 Overcurrent Protection Verification

The software overcurrent protection was tested by progressively reducing the load resistance below the rated 10 Ω until the INA219 reported a bus current exceeding the 700 mA threshold. At a load of approximately 8.3 Ω (I_out ≈ 701 mA), the firmware correctly latched the PWM output to zero within one control loop period (100 ms), reducing the output voltage to near zero and preventing thermal overstress. A serial alarm message was simultaneously transmitted. Manual reset via the serial monitor successfully re-enabled the converter at the previously commanded duty cycle.

5. Conclusion and Future Work

5.1 Conclusion

This project successfully designed, constructed, and experimentally characterized a digitally controlled DC-DC buck converter using the ESP32 microcontroller, TC4420 gate driver, MTA30N06HD power MOSFET, and INA219 current sensor. The principal outcomes are as follows:

- The ideal volt-second balance $V_out = D \times V_in$ was experimentally verified across the 20–90% duty-cycle range, with measured output voltages deviating from the ideal by no more than 3.75%, fully accounted for by conduction-loss parasitics.
- Peak converter efficiency of 88% was achieved at 3.4 W output power and 25 kHz switching frequency, consistent with theoretical loss models.
- Output voltage ripple of 42 mV_{pp} at rated load was within design specification, confirming the adequacy of the selected LC filter.
- The firmware-based overcurrent protection reliably interrupted converter operation within one control loop period upon detecting an over-current condition, with correct manual-reset behaviour.
- Real-time voltage, current, power, and efficiency telemetry was demonstrated via the INA219-to-ESP32 I2C link and Arduino Serial Plotter interface.

The hardware and firmware artefacts are released as an open-source reference design on GitHub, providing a documented, reproducible baseline for subsequent research and coursework.

5.2 Future Work

Several extensions are planned to evolve this prototype toward a fully autonomous smart energy node:

- Closed-Loop PID Control: The current open-loop implementation will be replaced by a discrete-time PID regulator implemented in firmware. The INA219 output voltage measurement will serve as the feedback variable, enabling load-independent regulation with sub-100 ms settling time.
- Higher Switching Frequency: Increasing the switching frequency to 100–200 kHz (ESP32 LEDC supports up to 40 MHz) will reduce the LC filter volume significantly, enabling PCB miniaturisation.
- Synchronous Rectification: Replacing the SR1645 Schottky diode with a second MOSFET and complementary gate drive will eliminate diode conduction losses and potentially raise peak efficiency above 93%.

- IoT Cloud Monitoring: The ESP32 Wi-Fi stack will be leveraged to publish telemetry (voltage, current, efficiency, thermal data) to an MQTT broker, enabling remote monitoring dashboards via Node-RED or Grafana.
- AI-Based Fault Prediction: An on-device TensorFlow Lite model trained on normal operating signatures will perform real-time anomaly detection, triggering graduated protection responses — aligning this node with the SmartNode-PK edge-AI architecture.
- PCB Layout and EMI Compliance: The breadboard prototype will be migrated to a two-layer PCB with a dedicated ground plane, Kelvin sense connections for the shunt resistor, and a ferrite bead input filter to meet IEC 61000-3-2 Class D conducted EMI limits.
- Smart Grid Integration: Multiple converter nodes will be networked via MQTT to implement a distributed droop-controlled DC microgrid, evaluating load sharing, fault isolation, and black-start capability within the broader GridGuardian architecture.

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Source Code & Documentation: github.com/rajedmehmood/Digitally-Controlled-DC-DC-Buck-Converter